

IGCSE Math

Extended 0580

Chapter 1:

Numbers

Welcome to AnithUncommon's Sample Resources!

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In IGCSE Math Extended 0580, 1.1-1.5

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by (Mentor Name) to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in (Subject Name), join our nonprofit!

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This sample is made thanks to, [Vihaan Amin](#)

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- **Classify** any number into its specific set (Natural, Integer, Rational, Irrational).
- **Deconstruct numbers** into their prime "atoms" and find shared traits between numbers (HCF/LCM).
- **Manipulate** the "growth" and "decay" of numbers through powers and roots.
- **Translate** fluently between the three languages of proportion: Fractions, Decimals, and Percentages.

Reflective Questions to Consider:

The world of numbers is divided into different groups and sets, each with its own specific rules. Some numbers are "clean" and easy to measure, while others are "messy" and go on forever. Understanding how to sort these numbers is the first step to solving any real-world problem.

- In a Venn Diagram, the "Intersection" is where two groups share common ground. How can we use this to visualize things that belong to two different worlds at once, like a number that is both "Prime" and "Even"?
- Why do we use Percentages for sales at a store, but Fractions for recipes in a kitchen? To what extent does the "format" of a number change how we perceive its value?

- If a Rational number is a clean fraction, and an Irrational number like Pi never ends or repeats, what does that tell us about the limits of human measurement?

Syllabus Outline:

Unit 1: 1.1-1.5	
Tip! • You may encounter difficulties remembering the symbols for Sets or the difference between Factors and Multiples. The main thing is to understand how these classifications help you organize data. Think of a Factor as a "building block" inside the number, and a Multiple as the "tower" built by stacking that number. If you get stuck on ordering, draw a quick number line! Activities in Class: • Collaborative Set Sorting: Work with your mentor to place 20 different numbers into a 3-circle Venn Diagram (Primes, Multiples of 3, Square Numbers). • 10-minute Kahoot Quiz: Focus on rapid-fire identification of rational vs irrational numbers. • Mark a long number and practice placing fractions, roots, and negatives in the correct order.	
1.1 - Types of Number • Natural Numbers are your basic counting tools (1, 2, 3...). Think of these as the first numbers humans ever needed. • Integers expand this to include zero and the negative values (... , -2, -1, 0, 1, 2...). • Prime Numbers are the elite building blocks. They	Mentors Note's <i>Check your Primes:</i> Remember that 1 is excluded from your list because a prime must have exactly two distinct factors. 1 only

only have two factors: 1 and themselves. You cannot break them down further. 2 is special because it is the only even prime. • Square and Cube Numbers are the result of a number multiplying by itself once (Square) or twice (Cube). Think of these as "Perfect Shapes." • HCF (Highest Common Factor) is the largest number that goes into both values. • LCM (Lowest Common Multiple) is the first number that appears in the times tables of both. It is the "Meeting Point" of two patterns. • Rational vs. Irrational: Rational numbers can be written as a/b (fractions). Irrational numbers are "messy," they are decimals that go on forever without a repeating pattern, like the square root of 2 or Pi.	has one factor, so it doesn't count! HCF/LCM Strategy: Use the "Venn Diagram Method" for difficult numbers. Put the prime factors of both numbers into two circles. Your HCF is what you get when you multiply the overlap; your LCM is the product of every single number visible in the circles. Reciprocals: Keep it simple, this just means "1 divided by the number." If you have a fraction, just flip it upside down and you are done.
1.2 - Sets • The Universal Set (represented by a curly E) is the boundary containing all elements under consideration. • The Epsilon Symbol indicates membership. If x is in Set A, you use this symbol to show it belongs there.	Mentors Note's The Outside Zone: <i>Don't forget to look at the elements sitting outside the circles but still inside the rectangle. These are</i>

• Union (U) is the "inclusive" operator. It includes everything in Set A, everything in Set B, and everything in the overlap. It's like a party where everyone from two different friend groups is invited. • Intersection (n) includes only the elements found in both sets simultaneously. • Complement (A') consists of all elements in the Universal Set that are NOT in Set A.	<i>still part of your Universal set!</i> <i>Watch the notation: When you see $n(A)$, it is asking you for the "number of items" in the set, not the names of the items themselves.</i>
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1.3 - Powers and Roots • Powers represent repeated multiplication. 5 to the power of 3 means 5 is multiplying itself three times ($5 \times 5 \times 5$). • Roots are the inverse operation. A square root looks at a number like 49 and asks, "What number multiplied by itself makes this?" (The answer is 7). A cube root looks for a number multiplied by itself twice to reach the target, like 5 for 125.	Mentors Note's <i>Speed is key: You should aim to recognize square roots up to 225 and cubes up to 125 instantly without a calculator. Practice these like they are vocabulary words.</i> <i>Negative Roots: You cannot find the square root of a negative number in this course, but you CAN find the cube root of a negative. For example, the cube root of -8 is -2,</i>
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	because (-2) times (-2) times (-2) equals -8 .
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1.4 - Fractions, Decimals, and Percentages • Fractions show the exact division of a numerator by a denominator. • Decimals use a point to separate whole numbers from fractional parts based on powers of ten. • Percentages express values as a portion of 100. This is the "standardized" version that allows us to compare different scales easily. • Recurring Decimals have a digit or sequence of digits that repeat infinitely. You use a dot above the repeating numbers to indicate this.	Mentors Note's <i>Understanding the 'Why': It is much easier for you to compare 75% vs 80% than it is to compare $\frac{3}{4}$ vs $\frac{4}{5}$. Percentages make life simpler!</i> <i>The 9s Trick: Here is a shortcut, any single repeating digit decimal (like 0.777...) can be turned into a fraction by just putting that digit over 9 (making it 7 divided by 9).</i>
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1.5 - Ordering • The < and > symbols indicate which value is smaller or larger. • To order a list like [$\frac{3}{4}$, 0.8, 72%], you must convert them all into the same format. Converting everything to decimals is usually the most efficient method.	Mentors Note's <i>The Negative Trap: Be careful! It is easy to think -10 is "bigger" than -5 because 10 is a bigger number.</i>
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- $\frac{3}{4}$ becomes 0.75; 72% becomes 0.72. Once standardized, the order is clear: 0.72, 0.75, 0.8.

Think of it as temperature: -10 degrees is much colder (lower) than -5 degrees. Always check the number line!

Mentors' Sources:

- <https://znotes.org/caie/igcse/mathematics-0580/>
- <https://www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf>
- <https://www.physicsandmathstutor.com/maths-revision/>